

Subdividing an Edge of K_9 : A Disproof of Conjecture 00000003476

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Abstract

The source asserts that subdividing an edge decreases the Wiener index if and only if the betweenness of that edge is below a quarter of the number of vertices. In the complete graph K_9 every edge has betweenness $1 < 9/4$, yet subdividing an edge raises the Wiener index from 36 to 53. The graph, the subdivision, the Wiener index, the betweenness and the negation of the criterion are formalised in Lean. In fact subdivision never decreases the Wiener index of a connected graph. An earlier submission argued with K_5 but had none of these objects in Lean; its error is described below.

Definitions and reading of the source. Let G be a finite connected simple graph with n vertices and distance function d . Its Wiener index is $W(G) = \sum_{\{x,y\}} d(x,y)$, the sum over unordered pairs of vertices. Subdividing an edge $e = ab$ gives the graph G_e : the edge ab is removed and a new vertex w adjacent to exactly a and b is added. The betweenness of e is

$$B(e) = \sum_{\{s,t\}} \frac{\sigma_{st}(e)}{\sigma_{st}},$$

where σ_{st} is the number of shortest paths from s to t and $\sigma_{st}(e)$ the number of those through e . The source claims, for every such G and e ,

$$W(G_e) < W(G) \iff B(e) < \frac{n}{4}. \quad (1)$$

Theorem 1. *Let $G = K_9$ and let e be an edge. Then $B(e) = 1 < 9/4$, while $W(G) = 36$ and $W(G_e) = 53$. Hence (1) fails.*

Proof. Any two distinct vertices s, t of K_9 are adjacent, so the only shortest path between them is the edge st . It passes through e exactly when $\{s, t\} = e$. Thus $B(e) = 1$. All $\binom{9}{2} = 36$ distances equal 1, so $W(G) = 36$.

Let $e = ab$. The graph G_e has 10 vertices and is connected, so each of its $\binom{10}{2} = 45$ distances is at least 1. Already this gives $W(G_e) \geq 45 > 36$. Precisely, the pairs at distance 2 are $\{a, b\}$ and $\{w, x\}$ for the seven vertices $x \neq a, b$; all other pairs are adjacent. So $W(G_e) = 45 + 8 = 53$. $\square$

Proposition 2. *For every finite connected graph G and every edge e , $W(G_e) > W(G)$.*

Proof. Let $e = ab$ and let d' be the distance in G_e . A walk in G_e between two old vertices x, y yields a walk in G that is no longer: replace each passage a, w, b or b, w, a by the edge ab , and delete each passage a, w, a or b, w, b . Hence $d'(x, y) \geq d(x, y)$. The pairs containing w contribute a positive amount. Summing gives $W(G_e) > W(G)$. $\square$

So the left side of (1) never holds, and the criterion fails for every edge with $B(e) < n/4$.

Remark. The value of $B(e)$ depends on the convention only mildly. Summing over ordered pairs gives 2; omitting the pair of endpoints gives 0; normalised versions are smaller. All are below $9/4$, which is why K_9 is used and not a smaller complete graph. If the quarter is taken of the 10 vertices of G_e , the same inequalities hold. For the clause on bipartite graphs, $K_{7,7}$ has $n = 14$ and $B(e) = 1 + 2 \cdot \frac{6}{7} = \frac{19}{7} < \frac{14}{4}$ for every edge, so by the proposition the criterion fails there as well. The proposition and this last computation are not formalised.

The earlier submission and its error

Pull request 225 of the competition repository (by `orionsheep`) proposed the same kind of counterexample with K_5 . It was closed without merging on 3 October 2026. The reviewer’s stated reason was that the main Lean theorem was arithmetic on a hard-coded list of distances, and that no graph, Wiener index, subdivision or betweenness was defined in Lean.

That is the error: the Lean file checked that a list of numbers sums to 19 and that $19 > 10$, which says nothing about graphs. There is also a gap in the example itself. With betweenness summed over ordered pairs an edge of K_5 has betweenness $2 > 5/4$, so K_5 refutes the criterion only under some conventions.

The present submission defines the four objects in Lean for arbitrary finite graphs, states the criterion with a universal quantifier, and uses K_9 , for which the ordered convention is covered too.

Formal verification and scope

Graphs are Mathlib’s `SimpleGraph` and distances are Mathlib’s `SimpleGraph.dist`. `subdivide G a b` is a graph on `Option V`; the new vertex is `none`. `wiener` is half the sum of distances over ordered pairs. `sigma` and `sigmaThrough` count the walks whose length equals the distance, and those among them whose edge list contains e ; `betweenness` is the resulting sum over unordered pairs and `betweennessOrdered` the sum over ordered pairs.

For a complete graph on any vertex type, `sigma_top` and `sigmaThrough_top` prove $\sigma_{st} = 1$ and $\sigma_{st}(e) \in \{0, 1\}$ according to whether $\{s, t\} = e$, by classifying all walks of length one. This gives `betweenness_K9`: $B(e) = 1$, and `betweennessOrdered_K9`: the ordered sum is 2. `wienerTwice_K9` computes the ordered distance sum 72. `subdivide_K9_connected` proves that the subdivided graph is connected, and `wienerTwice_ge` proves that a connected graph on m vertices has ordered distance sum at least $m(m - 1)$. Hence `wiener_increases`: $W(K_9) < W((K_9)_e)$. Lean proves the bound $W((K_9)_e) \geq 45$, which is what the disproof needs; the exact value 53 is the hand computation above.

The proposition `ClaimedCriterion` is (1) quantified over all finite connected graphs and edges. Its negation is the theorem

`conjecture_false.`

The theorem

`conjecture_false_ordered`

is the same for the ordered convention. No `sorry`, `native_decide` or custom axiom occurs.

The disproof concerns the stated equivalence. No claim is made about the phrase on the cut-load inequality chain, which is not a precise statement.

Source and reproduction

The exact bilingual source is included byte-for-byte as `SOURCE.md`. Its repository path is

`conjectures/00000003476.md.`

The project pins Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b.`

From `lean/`, run `lake build`, then

`lake env lean -DwarningAsError=true Main.lean.`

The included public-Git manifest locks all dependency revisions. The `verification/` directory records the fresh-build logs, theorem-axiom reports, artifact hashes, review and final PDF inspection. No supplementary numerical computation is required.